

Revision Notes for Class 7 Maths Chapter 7 A Tale of Three Intersecting Lines

Triangles are some of the simplest and most important shapes studied in mathematics. Every triangle has three vertices (corner points) and three sides (line segments joining the vertices). The points are usually named with capital letters such as A, B, and C, and the triangle is written as $\triangle ABC$.

Triangles appear in different forms. The symbol $\triangle$ is used to represent a triangle, and any three non-collinear points can determine a unique triangle. If all three points are on a straight line, a triangle cannot be formed.

Angles in a Triangle

In any triangle, the meeting points of sides form three angles. For triangle $\triangle ABC$, the angles are denoted as $\angle A$, $\angle B$, and $\angle C$. Angles play an important role in classifying and constructing triangles.

Equilateral Triangles and Compass Construction

An equilateral triangle has all three sides of equal length. It is the most symmetric type of triangle. To construct an equilateral triangle of side 4 cm accurately, a ruler and compass are used. First, draw a line $AB = 4$ cm. Then, keeping the compass at A, draw an arc with a radius of 4 cm. Repeat this from B. The intersection of the two arcs is point C. Joining AC and BC completes the equilateral triangle.

Constructing a Triangle with Given Sides

To draw a triangle with given side lengths (e.g., 4 cm, 5 cm, and 6 cm), use a compass for accuracy. For example, draw $AB = 4$ cm. With A as the center, draw an arc with a radius of 5 cm. With B as the center, draw another arc of 6 cm. Mark C at the intersection of arcs, then join AC and BC.

- Equilateral triangle: all three sides equal.

- Isosceles triangle: two sides equal.
- Scalene triangle: all three sides different.

Triangle Inequality and Side Lengths

Not all sets of three measurements can form a triangle. The rule is that the sum of the lengths of any two sides must be greater than the third side. This is called the triangle inequality. If even one side is equal to or greater than the sum of the other two, a triangle cannot exist.

- If you try side lengths 3 cm, 4 cm, 8 cm, you cannot make a triangle because $3 + 4 = 7$ is less than 8.
- For sides 4 cm, 5 cm, and 8 cm, because $4 + 5 = 9 > 8$ and other combinations work too, a triangle can be formed.

There are three cases in triangle construction using circles (representing compass arcs): (1) Arcs touch each other exactly (triangle with colinear points), (2) Arcs do not intersect (impossible triangle), (3) Arcs intersect at two points (triangle can be constructed). Only the third case gives a valid triangle.

Constructing Triangles with Angles and Sides

Sometimes, a triangle is given by two sides and the included angle (SAS condition). For example, if $AB = 5$ cm, $AC = 4$ cm, and $\angle A = 45^\circ$, first draw AB. At A, construct a 45° angle, measure and mark $AF = 4$ cm, and connect FB.

Other times, a triangle is defined by a side and two angles (ASA condition). For instance, draw $AB = 5$ cm as base. At A and at B, construct the given angles. Where the arms meet is point C. The triangle is completed by connecting all sides.

- If the sum of the two given angles is less than 180° , construction is possible. The third angle is always 180° minus the sum of the other two.

- If both angles are right or obtuse (total sum $\geq 180^\circ$), it is impossible to draw a triangle.

Angle Sum and Properties

The sum of all three interior angles in any triangle is always 180° . This is known as the angle sum property. This can be seen by drawing a line parallel to one side of the triangle and using alternate and interior angles or by folding a paper cut-out.

An exterior angle of a triangle (formed by extending a side) is always equal to the sum of the two opposite interior angles. For example, in $\triangle ABC$, if $\angle A = 50^\circ$ and $\angle B = 60^\circ$, the exterior angle at C is 110° because $180^\circ - \angle C = \angle A + \angle B$.

Altitudes of a Triangle

An altitude of a triangle is a perpendicular drawn from a vertex to its opposite side (or to the side produced). Every triangle has three altitudes, and they can meet inside or outside the triangle depending on its type. In a right-angled triangle, one altitude coincides with a side.

- To draw an altitude from vertex A to side BC, align a ruler along BC, then use a set square to draw a perpendicular line from A to BC.

For obtuse triangles, some altitudes may have to be constructed by extending the side beyond the triangle. Altitudes are important for finding the height and area of triangles.

Classification of Triangles

Triangles are classified based on the length of sides and on the angles within them. Based on sides, they can be:

- **Equilateral:** Three sides equal, three equal angles (each 60°).
- **Isosceles:** Two sides equal and two angles equal.

- **Scalene:** All sides and all angles are different.

Based on angles, triangles are:

- **Acute-angled:** All angles are less than 90° .
- **Right-angled:** Has one angle exactly 90° .
- **Obtuse-angled:** Has one angle greater than 90° but less than 180° .

It is not possible to construct a triangle that is both equilateral and right-angled or obtuse-angled, but an isosceles triangle may be right-angled or obtuse-angled depending on the sides and included angles.

Summary of Key Points

- Use of compass and ruler improves accuracy in triangle construction.
- Triangle inequality must be checked before starting any construction.
- Triangles can be constructed with:
 - Three sides (if triangle inequality holds)
 - Two sides and included angle
 - A side and two angles (sum $< 180^\circ$)
- The sum of all angles in any triangle is always 180° .
- Each triangle has three altitudes which may or may not fall inside it, depending on its type.

Triangles are fundamental in geometry and understanding their properties helps in solving various real-life mathematical and construction problems.